

All questions are mapped to [CLO-1, C3].

1. Association rule learning is mainly used to: [1]
- Predict continuous values ✗
 - ✓ Discover relationships between variables
 - Detect anomalies ✗
 - Reduce feature dimensions ✗
2. In k-fold cross-validation, each data point is used: [1]
- Only once for training
 - ✓ Only once for validation
 - k times for training
 - k-1 times for validation
3. Which of the following is an example of systematic noise? [1]
- Random sensor fluctuations
 - Random label assignment ✗
 - ✓ A thermometer that always shows lower temperature than actual
 - Typographical errors during data entry ✗
4. In a contour plot, the direction of steepest descent is: [1]
- Parallel to contour lines ✗
 - ✓ Perpendicular to contour lines (ideal)
 - Random
 - Along the x-axis
5. Which of the following best describes *batch size*? [1]
- Number of epochs used for training ✗
 - Number of features in the dataset ✗
 - ✓ Number of samples processed before updating model parameters
 - Number of iterations in one epoch ✗
6. Feature scaling is important in gradient descent because it: [1]
- Improves model accuracy ✗
 - Makes the cost function convex ✗
 - Speeds up convergence ✗
 - ✓ Removes outliers

7. A company collects 5,000 customer profiles with purchase behavior but no predefined labels. Identify the most suitable learning paradigm and sub-type if any. [2]

unsupervised; clustering learning

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8. A vocabulary contains 5,000 words. If word embeddings of size 300 are used instead, how many values represent one word? [1]

vocabulary = 5000 words.

$\frac{5000}{300} \Rightarrow 16.67$ values represent one word.

9. A dataset is evaluated using 10-fold cross-validation. What fraction of data is used for training in each fold? [1]

$\frac{9}{10}$ for training ~~that is~~ folds

Here, $k=10$ so 10 segments out of which 9 will be ^{for} training.
~~means 10 folds.~~

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